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DOCUMENT NO. IPS-MBD22927-ME-1432

SPECIFICATION

FOR

SCISSOR LIFT

RESonate, Jhagadia, Gujarat

Rev	Date	Description	Made by	Chkd by	App by
R0	July 24, 2023	For Enquiry	ASP	PDP	PDP

☐	Entire Specification Issued this Revision	SPECIFICATION ISSUED FOR			
		☐	In-house Review and replies	☐	Purchase
☐	Revised Pages Only Issued this Revision	☐	Client Approval	☐	Construction
		☒	Enquiry	☐	Engineering

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1.0 Project Information

Thermax Ltd is planning to set up a greenfield Ion Exchange Resin manufacturing facility at Jhagadia, Gujarat.

There is existing plant (unit-1) adjacent to Unit-2. Unit- 2 is being considered for recent expansion. Unit-2 plot area is 67000 m2. dedicated to this plant. Unit-1 and Unit-2 will be amalgamated to make additional area available for tank farm, non-plant building and other structure as per space availability.

Plant capacity will be 18,000 m^3. There will be 3 independent product plants within the single building. Products are Polymers, Cations and Anions. Utilities and other requirements to be planned for 18,000 m^3.

2.0 Project Details

1.	Owner	:	Thermax Ltd
2.	Project	:	RESonate
3.	Owner's representative	:	IPS-Mehtalia Pvt. Ltd.
4.	Location	:	Jhagadia
5.	Latitude	:	21.6393 Degree North
6.	Longitude	:	73.1187 Degree East
7.	Min. dry bulb temperature	:	6.9 °C
8.	Max. dry bulb temperature	:	44.2 °C
9.	Min. wet bulb temperature	:	4.8 °C
10.	Max. wet bulb temperature	:	30.3 °C
11.	Min. RH	:	37.1 %
12.	Max. RH	:	72.8 %
13.	Site elevation from mean sea level	:	15 M. (of Bharuch)
14.	Basic wind speed	:	44 mtr/s
15.	Seismic zone	:	III
16.	Nearest city	:	Bharuch, Ankleshwar
17.	Nearest railway station	:	Bharuch, Ankleshwar
18.	Nearest airport	:	Surat International Airport Vadodara Airport
19.	Nearest port	:	Dahej Port

3.0 Scope

The scope of specification covers design, fabrication, assembly, inspection, supply, packing and forwarding, transportation, transit insurance, installation, testing with load and obtaining permissions if any for operating the scissor lift from Statutory authorities.

Sr. No.	Tag No.	Quantity (Nos.)	Capacity (Kgs)
1.	Lift 01	1 No.	1000

4.0 Technical Requirement

4.1 Design Standards

Sr. No.	Description		Design Code / Standard No.
1.	Safety Standard		EN 292 / EN 1570
2.	For electric motor	:	Inverter duty motor as per IS - 325 – 1996 and Energy efficiency type IE3 as per IEEMA Std. /19-2000 IP 55 to Flame proof, Confirming to Zone 1 (IS 5572) Gr II A & II B Vapours (IS 2148)
3.	Factor of Safety	:	As per IS Standard
4.	Earthing	:	IS: 3043 – 1987
5.	Instrument Type	:	All instrument components shall be flame proof type

4.2 Safety Requirement

1. Scissor lift shall be designed for operation using manual forklift or other Material handling equipment. Lift shall be designed for point loads of manual forklift along with load on it. Limit switch for door shall be provided and interlock shall be provided so that door where lift is landing shall only open.
2. Beneath Lift, MS plate shall be provided on one side to cover pit section when lifted.
3. All Flameproof components shall be terminated in Flameproof Box with Proper gland packing
4. Noise level shall be less than 75dB.
5. No asbestos material shall be used anywhere in the system.
6. Equipment shall be designed for the ease of maintenance cleaning and inspection.
7. Power pack shall be kept in pit itself.
8. Lift shall have load cell to detect the overload capacity and accordingly provide an indication in the hall panel with an audio alarm.
9. All power cables shall be XLPE copper armored cable only.
10. Anchors point for safety bolt while working on top of the chain for maintenance.

11. Safety railing shall be provided on car on 3 sides of car to avoid fall during maintenance activities. Railing height shall be 1.2 meters and 0.4m meters from bottom.

5.0 Utilities Available

Sr. No.	Description		Utility
1.	Power supply	:	415 ±10%, 3 phase, 50±3%Hz.

6.0 Battery Limit

6.1 Electrical

a. Single point 3-phase, 4 wire, power termination by client and cable gland by vendor.
b. Single point earthing connection.

6.2 Civil

a. All civil construction for Lift well (in RCC/Brick) will be by client.

7.0 Vendor's scope

7.1 Equipment with all accessories as specified in technical requirements.
7.2 Lubricants parts used should be food grade.
7.3 Transportation, unloading at site, locking and safety of equipment stored on site. Installation and testing of lift along with all commissioning consumables except utilities such as water, power etc.
7.4 Special tools and tackles required for installation and testing
7.5 Client will give only input power to panel 3 phase 415V, rest all work in vendor's scope.
7.6 Earthing at single point only. Further distribution in vendor's scope.

8.0 Exclusions

8.1 All civil work.
8.2 Laying and termination of power cables to control panel.
8.3 Material storage area.
8.4 Provision of power supply.
8.5 Lighting and bulkhead fittings in well and machine-room.

9.0 Inspection and testing

9.1 All equipment and their accessories shall be subjected to inspection as per approved QAP.
9.2 No equipment or its accessories / part shall be shipped without testing at shop up to full satisfaction of inspection authority.
9.3 Performance test of equipment shall be carried out at site as per relevant acceptance standard.
9.4 Vendor should maintain internal inspection records at all times, if any stages are found lacking by way of proper records, inspection authority reserves the right to ask vendor to modify / amend the fabrication stage.
9.5 Any such inspection at vendor's work or inspection waiver however shall not relieve vendor from his responsibility for supply of equipment / its accessories as per specifications and other obligations stipulated in contract.

10.0 Erection and commissioning

10.1 Vendor to do erection and commissioning of the equipment.

10.2 Vendor / vendor's representative shall consider 2 pre-commissioning visits at site after material dispatch. One for ensuring delivery of all material and condition of material delivered. In case of incomplete supply or any damage to the material during transit, immediate action should be taken by vendor. Such replacement and transportation and other charges for material missed in the lot shall be done by vendor at free of cost.

10.3 Vendor should arrange required manpower (skilled and unskilled) and tools, tackles required for erection and commissioning, and shall not borrow the same from other agency in operation at site.

10.4 Vendor shall ensure that erection and commissioning is carried by his personnel under safety norms and other obligations applicable such as ESIC, WCP. PF etc.

10.5 Load trial shall be conducted by vendor after installation the lift.

11.0 Warranty and penalty

11.1 Complete Equipment / system shall be warranted for trouble free operation for the period of 12 months from the date of commissioning or 18 months from the date of receipt at site whichever is earlier.

11.2 All components shall be warranted against guaranteed against faulty design, defective or improper materials poor workmanship or failure from normal use during the warrantee period.

11.3 In case of non-fulfilment of warrantee, vendor shall repair / replace such defective part/parts of the equipment supplied. Vendor shall be liable for all expenses arisen for such repair / replacement.

12.0 Post Order Support

Support shall be extended to the client for

1. Operation and maintenance training

13.0 Document submittal

13.1 Along with offer (Soft Copy)

1. Enquiry specification with acceptance / deviation marked along with filled technical datasheets. All pages shall be dully stamped and signed by vendor (in PDF/ Excel sheet)
2. GA drawing confirming the adequacy of the available well size, machine room, heights etc. (in AutoCAD/PDF)

13.2 After order for approval within 1 week of LOI/PO (in Soft Copy/Hard Copy (2 Sets))

1. General Arrangement Drawing and foundation drawing showing overall dimensions (in AutoCAD/PDF).

13.3 Before commissioning of system. (In Soft Copy/Hard Copy (2 Sets))

1. GAD and equipment layout showing overall dimensions (in AutoCAD/PDF).
2. Material test certificate / manufacturers test certificate for all components. (PDF)
3. Flame proof certificate for all FLP components (PDF)

4. Valid calibration certificate for all instruments (PDF)
5. Operation and maintenance manual (PDF)

14.0 List of Abbreviation

Sr. No.	Abbreviation		Name of Abbreviation
1.	*	:	Vendor to specify
2.	FAT	:	Factory acceptance test.
3.	LOI	:	Letter of intent.
4.	PO	:	Purchase order.
5.	GAD	:	General Arrangement Drawing
6.	VFD	:	Variable Frequency Drive

15.0 List of Annexure

Sr.No.	Description	Document No.	Revision No.
1.	Datasheet for Scissor lift	IPS-MBD22927-ME-1432	R0

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